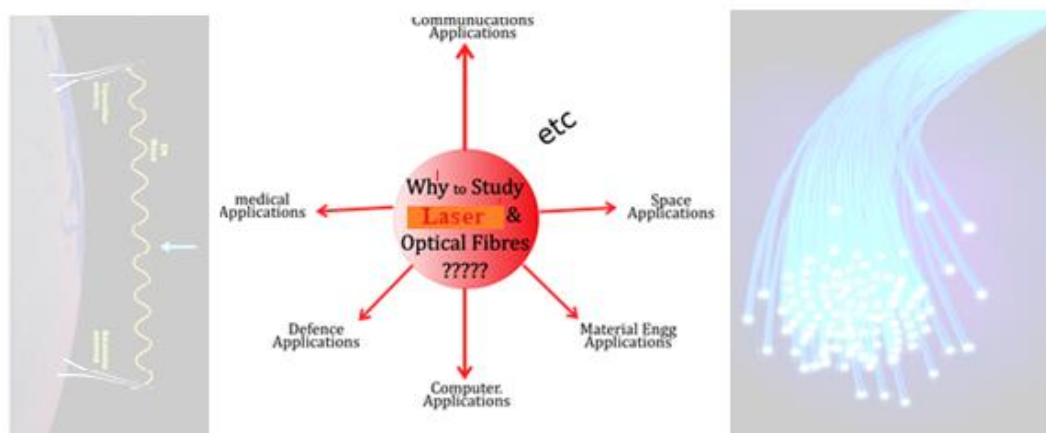


Module-I**(CSE Stream)****LASERS and OPTICAL FIBERS****Syllabus**

LASER: Characteristic properties of a LASER beam,, Interaction of Radiation with Matter, Einstein's A and B Coefficients, and Expression for Energy Density (Derivation), Laser Action, Population Inversion, Metastable State, Requisites of a laser system, Semiconductor Diode Laser, Applications: Bar code scanner, Laser Printer, Laser Cooling(Qualitative), Numerical Problems.

Optical Fibers: Principle and structure, Propagation of Light, Acceptance angle and Numerical Aperture (NA), Derivation of Expression for NA, Modes of Propagation, RI Profile, Classification of Optical Fibers, Attenuation and Fiber Losses, Applications: Fiber Optic networking, Fiber Optic Communication. Numerical Problems

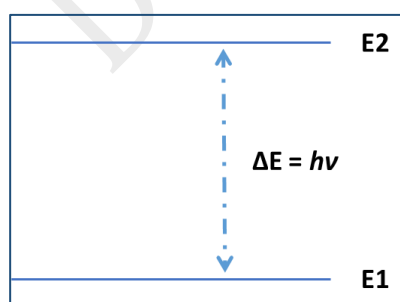


The word **LASER** is an acronym for **Light Amplification by Stimulated Emission of Radiation**. Its working principle depends on the phenomenon of stimulated emission, first predicted by Einstein in 1916.

Characteristic Properties of Laser: Highly Monochromatic, High Directionality, Highly coherent and extremely intense.

Interaction of Radiation with Matter:

Radiation interacts with matter under appropriate conditions. The interaction leads to an



abrupt transition of the quantum system such as an atom or a molecule from one energy state to another. If the transition is from a higher state to a lower one, the system emits a part of its energy and if the transition is in the reverse direction, then it absorbs the incident energy.

In order understand the manner in which radiation interact with matter, consider two energy states of a system (E_1 & E_2). If the energy

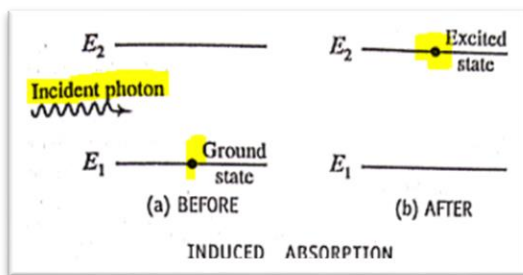
difference between the two energy levels is ΔE , then $\Delta E = (E_2 - E_1)$

Max Planck suggested that if an electromagnetic radiation of energy $h\nu$ incident on the system which is in the lower state E_1 , then the system can move to the higher state E_2 by absorbing the energy; $\Delta E = h\nu = (E_2 - E_1)$. On the other hand, if a system in state E_2 emits a radiation (a photon) of frequency ν , then system energy changes to E_1 .

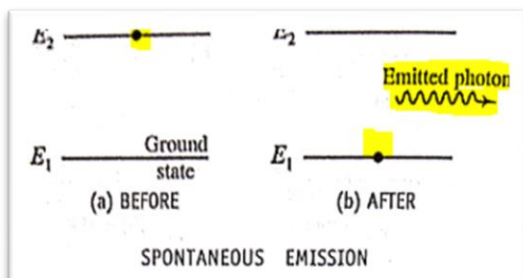
There are three possible ways through which interaction of radiation with matter can take place. Absorption of radiation is one, known as **induced absorption**. But the emission, as suggested by Einstein (1917), can occur in two ways; **spontaneous emission** and **stimulated emission**.

Spontaneous and Stimulated Processes:

1. Induced (stimulated) Absorption: It is the absorption of an incident photon by a system, because of which the system is elevated from a lower energy state (E_1) to higher state (E_2), wherein the difference in energy of two states (ΔE) is precisely the energy of photon ($h\nu$).

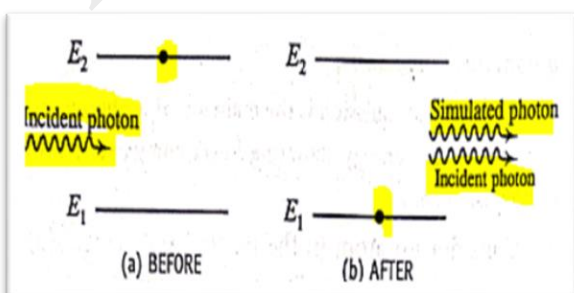


2. Spontaneous Emission: It is the emission of photon, when a system transits from a higher state (E_2), to lower energy state (E_1) without the aid of any external energy.



3. Stimulated Emission: It is the emission of a photon by a system, under the influence of an external photon of suitable energy, due to which the system transits from higher state (E_2) to a lower energy state (E_1). The photon thus emitted is called **stimulated photon**.

This kind of emission is responsible for 'laser action'



Einstein coefficients: (Expression for energy density):-

Consider two energy states E_1 and E_2 of a system of atoms ($E_1 < E_2$), having N_1 and N_2 number of atoms per unit volume, respectively. Let there be radiations of incident energy such that $h\nu = (E_1 - E_2)$. Let U_ν be energy density of radiations of frequencies in the range ν and $\nu + d\nu$ at temperature T .

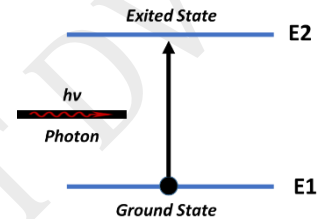
Let us now consider the induced absorption, two emission processes as follow,

(i) Case of Induced Absorption: In this case, an atom in the level E_1 can go to the level E_2 , when it absorbs a radiation of energy $h\nu = (E_1 - E_2)$. The number of such absorptions per unit time per unit volume is called **rate of absorption**.

Since, rate of absorption is proportional to number density N_1 and also to U_ν .

$$\rightarrow \text{Rate of absorption} \propto N_1 U_\nu = B_{12} N_1 U_\nu \quad (1)$$

where, B_{12} is proportionality constant or **Einstein coefficient of induced absorption**.

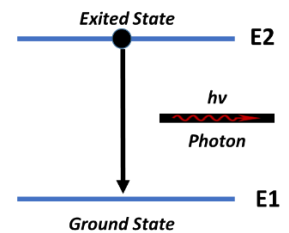


(ii) Case of Spontaneous Emission: In this case, an atom in the higher level E_2 undergoes transition to the lower energy level E_1 voluntarily by emitting a photon (see Fig). The number of such spontaneous emission per unit time per unit volume is called **rate of spontaneous emission**.

Since, spontaneous emission is proportional to only number density N_2 .

$$\rightarrow \text{Rate of spontaneous emission} \propto N_2 = A_{21} N_2 \quad (2)$$

Here, A_{21} is proportionality constant or **Einstein coefficient of spontaneous emission**.

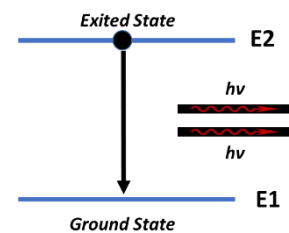


(iii) Case of Stimulated Emission: Here, the system requires an external photon of appropriate energy, $h\nu = (E_1 - E_2)$ to stimulate an atom in E_2 to the level E_1 and thereby cause stimulated emission. The number of such stimulated emission per unit time per unit volume is called **rate of stimulated emission**.

Since, the rate of stimulated emission is proportional to N_2 and also to U_ν .

$$\rightarrow \text{Rate of stimulated emission} \propto N_2 U_\nu = B_{21} N_2 U_\nu \quad (3)$$

here, B_{21} is proportionality constant or **Einstein coefficient of stimulated emission**.



At thermal equilibrium, No. of photons absorbed by system per sec = No. of photons emitted per sec.

$\therefore$ **Rate of absorption = Rate of spontaneous emission + Rate of stimulated emission**

$$\text{i.e., } B_{12} N_1 U_\nu = A_{21} N_2 + B_{21} N_2 U_\nu \quad \left| \text{using Eqns. (1), (2) \& (3)} \right.$$

$$\text{or, } U_\nu (B_{12} N_1 - B_{21} N_2) = A_{21} N_2 \Rightarrow U_\nu = \frac{A_{21} N_2}{B_{12} N_1 - B_{21} N_2}$$

$$\text{On rearranging, } U_\nu = \frac{A_{21}}{B_{21}} \left[\frac{1}{\frac{N_1 (B_{12})}{N_2 (B_{21})} - 1} \right] \quad (4)$$

According to **Boltzmann's law**, we have,

$$\frac{N_1}{N_2} = \frac{e^{-E_1/KT}}{e^{-E_2/KT}} = e^{(E_2-E_1)/KT} \quad \text{or} \quad \frac{N_1}{N_2} = e^{hv/KT} \quad (5) \quad |\because (E_2 - E_1) = hv$$

Putting **eqn. (5)**, in **(4)**, we have,

$$U_v = \frac{A_{21}}{B_{21}} \left[\frac{1}{e^{hv/KT} \left(\frac{B_{12}}{B_{21}} \right) - 1} \right] \quad (6)$$

Since, it is well known that; $B_{12} = B_{21}$

This implies that the probability of induced absorption is equal to the probability of stimulated emission. Because of such identity, the subscripts could be dropped and A_{21} & B_{21} can be represented simply as **A** & **B** and **eqn. (6)** rewritten as,

$$U_v = \frac{A}{B[e^{hv/KT} - 1]}$$

This is the expression for **energy density** in terms of **Einstein coefficients**.

Production of Lasers:

Some Important terms: **Pumping:** It is the process of exciting atoms from lower energy state to a higher energy state by supplying energy from an external source.

It can be achieved by different methods: (i) Excitation by a strong source of light (**optical pumping**) (ii) by electron impact (**electrical pumping**) (iii) by chemical reaction (**chemical pumping**) etc.,

Lasing: It is the process, which leads to emission of stimulated photons after establishing the population inversion.

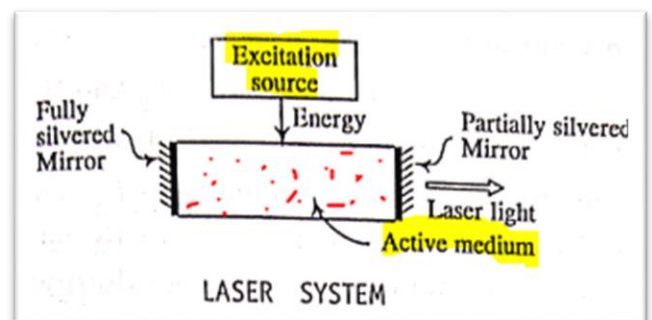
Active Medium: It is the quantum system in which, the pumping and the lasing action takes place

Laser cavity or resonant cavity or optical resonator: An active medium bound between two mirror forms a resonant cavity and referred as **optical resonator**.

Requisites of a Laser System:

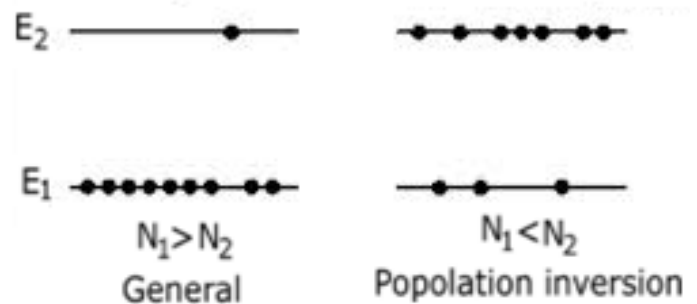
The basic structure of every laser device consists of,

- 1. Energy source** for pumping action: The excitation source provides energy in an appropriate form for pumping the atoms to higher energy levels.
- 2. Active medium** to support population inversion: A part of the input energy is absorbed by the active medium in which population inversion occurs and hence the medium attains capability to issue laser light.
- 3. A laser cavity (optical resonator):** The laser cavity provides necessary feedback (a part of output energy fed back into the medium).

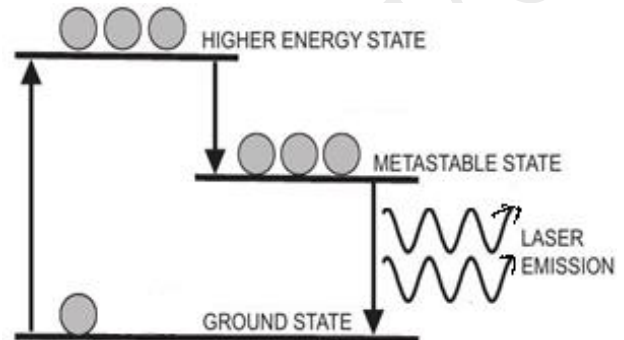


Conditions for Laser Action:**Population Inversion:**

It is the state of a system at which, the population of atoms in the higher energy state is more than that of lower energy state.

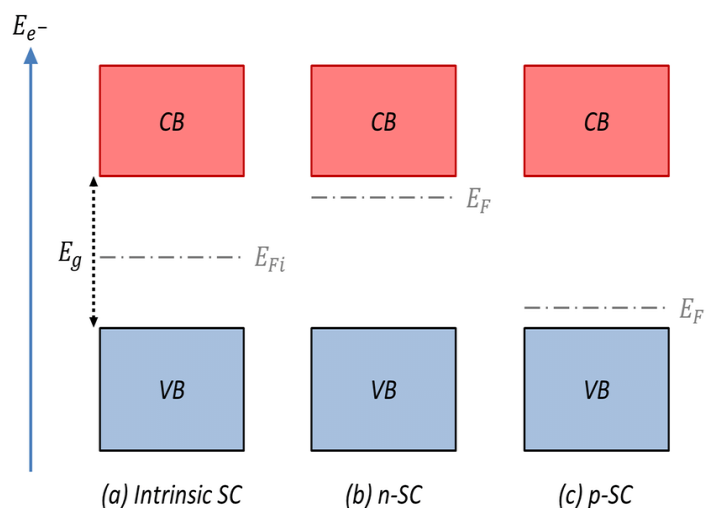
**Meta Stable State:**

Population inversion conditions do not exist under normal conditions. However, it is possible to achieve in systems, which possess a special kind of excited energy states called **Metastable States**. A metastable state differs from any ordinary excited state from its longer lifetime. Ordinary excited state has lifetime of 10^{-8} sec while, the metastable state possess lifetime of the order of 10^{-3} to 10^{-2} sec. This property helps in achieving population inversion and hence, the laser action.

**Semiconductor Diode Laser**

Semiconductor laser is the one in which the active medium is formulated by semiconducting material. The first semiconductor laser was made from GaAs, a homo junction semiconductor laser.

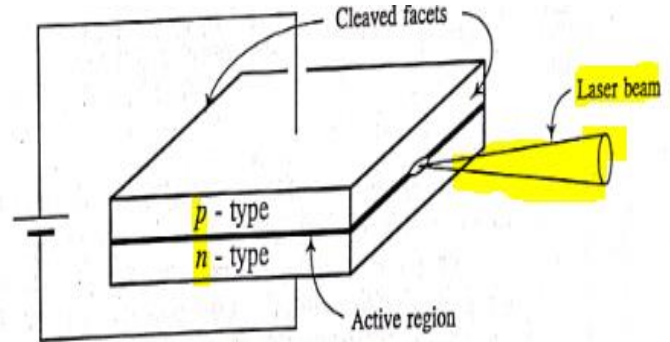
Degenerate semiconductors: They are semiconductors with very high doping. **Fermi level is the highest occupied energy state of the system at 0K.** In intrinsic semiconductor Fermi level lies exactly in the middle of the energy gap. But when the doping concentration increases, the Fermi level alters its position depending on the doping concentration. In degenerate n-type semiconductor, Fermi level lies within the conduction band and in degenerate p-type semiconductor Fermi level lies within the valence band.



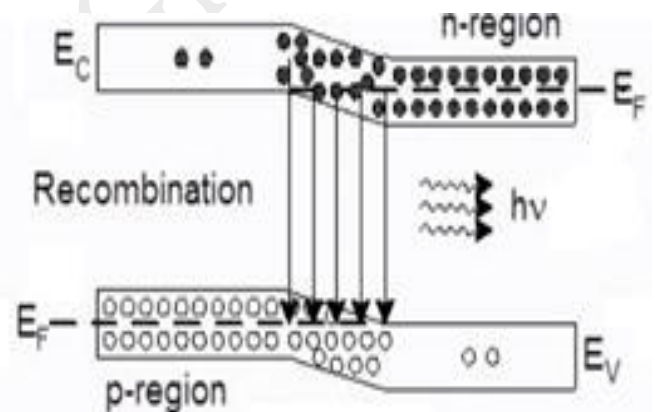
Principle: It works on the principle of stimulated emission. In a semiconductor laser, the population inversion is achieved by driving current. When current surpasses a threshold value, population inversion takes place and stimulated emission overtakes the spontaneous emission thus leading lasing.

Construction:

- A typical semiconductor laser is shown in **Fig.** It consists of heavily doped **n**- region and doped **p**- regions - **Active medium**
- Two parallel planes perpendicular to the junction are cleaved/polished, so that they act as reflecting mirrors and other two surfaces roughened to suppress reflections of the photons – **Optical Resonator**
- The end surfaces of the **p**- and **n**- sections parallel to the plane of the junction are provided with electrodes (for **electrical pumping**) in order to facilitate application of a forward bias voltage with the help of a voltage source.

**Working:**

- When a suitable forward bias is applied to the laser diode, electrons from **n**-side and holes from **p**-sides are injected into the junction, as a result current begins to flow.
- These charge carriers recombined and photons are emitted. This is *spontaneous emission*, which occurs in all the directions at low current.
- As the voltage is gradually increased due to forward biasing, **population inversion** is achieved between the valence band and conduction band, at which the emitted photons stimulate the emission of more photons.
- These photons are internally reflected several times at the polished walls, stimulating more and more photons, all coherent with them. When the photon beam becomes sufficiently intense, it emerges out from the junction as **laser**.
- Note: Typical Ex: The energy gap of GaAs is 1.4 eV, the wavelength of the laser beam emitted is,



$$\lambda = \frac{hc}{E_g} = 840 \text{ nm} \quad (\text{IR region}).$$

Applications: Semiconductor lasers are small and least expensive, can be produced in bulk. It finds application in optical communications as they provide light beams of wavelengths with low absorption loss in optical fibers.

Applications of Lasers:

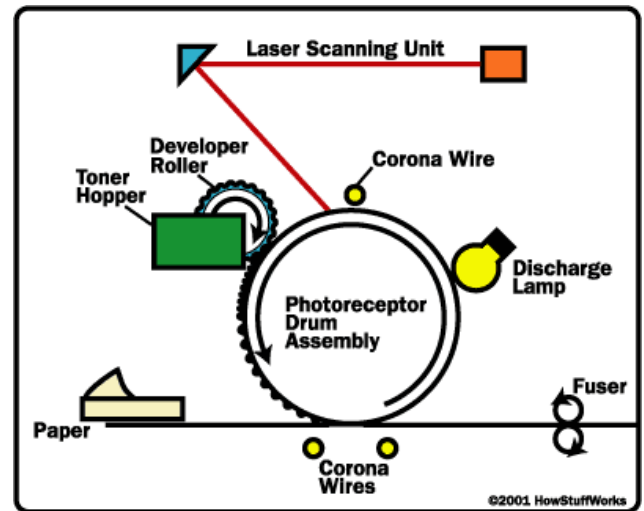
Due to unique properties of laser, it finds remarkable applications in a diversity of fields such as defense, engineering, medicine, material processing, communications, energy resources, 3-D photography (Holography), etc. Some important applications are discussed as follows;

LASER Printer:

Laser printers are digital printing devices used to create high quality text and graphics on plain paper. Their principle of operation involves electrophotography, also called xerography. A diode laser is used in the process of printing.

Construction and Working:

- A LASER beam projects an image of the page to be printed onto an electrically charged rotating photoreceptive/photosensitive drum coated with selenium.
- Photo conductivity allows charge to escape from areas exposed to laser light, and area gets positively charged. This can be done by the corona wire which charges up the photoreceptor drum so the drum gains a positive charge spread uniformly across its surface.
- The toner is then electrostatically attracted to the positively charged regions of the drum.
- The drum then prints the image onto paper by direct contact and heat, which fuses the ink to the paper.

**Advantages:**

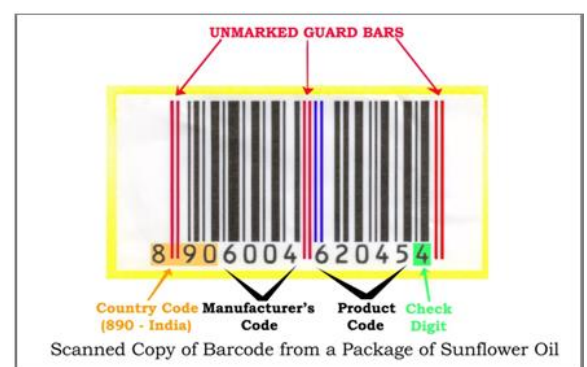
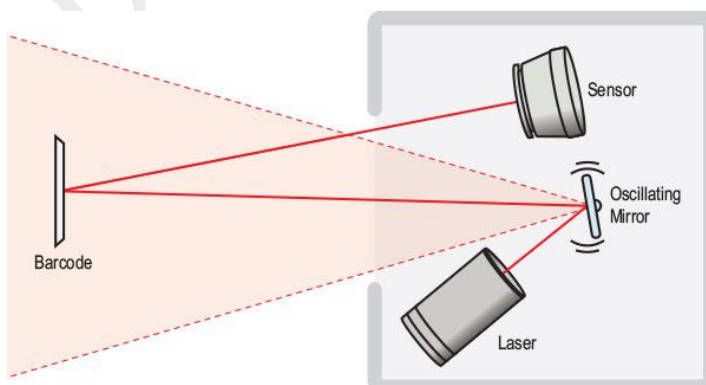
- ✓ Laser printers are generally quiet and fast.
- ✓ Laser printers can produce high quality output on ordinary papers.
- ✓ The cost per page of toner cartridges is lower than other printers.

Disadvantages:

- ✓ The initial cost of laser printers can be high compared to dot-matrix printers and ink-jet printers.

LASER Bar-Code Scanner:

Bar code scanners read the Universal Product Code printed on most items in supermarkets and other stores. The scanner reads the code automatically, looks up the appropriate price in memory, and prints a detailed sales slip. A barcode is used to encode information in a visual pattern readable by a machine.

Construction and Working:

- It mainly consists of three different parts; a Laser Light source, Oscillating mirrors and Sensor/Detector.
- A low power Laser beam is used to read the code.
- The Laser beam is moved across the universal code using Oscillating mirrors.
- As the item is moved across the Laser beam, dark areas of the code absorb the light and the bright areas reflect it.
- The presence or absence of the reflected light is registered by the photosensitive light detector as analog signal.
- Analog signal is then sent to the decoder. The decoder interprets that signal, validates the barcode using the check digit, and converts it into text. This converted text is delivered by the scanner to a computer system holding a database of the maker, cost, and quantity of all products sold.

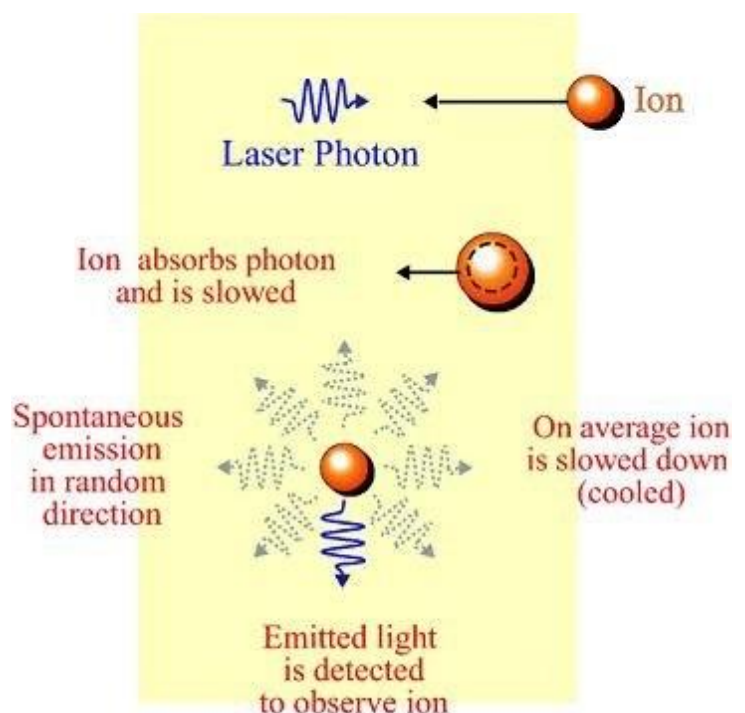
LASER Cooling:

Laser cooling is a multi-process that includes a number of techniques in which atomic and molecular samples are cooled down to a temperature near absolute zero. These techniques depend on the fact that when an object, which is usually an atom, re-emits a photon - a particle of light - its momentum is subject to change.

Principle of LASER Cooling: Laser cooling is the use of dissipative light forces for reducing the random motion and thus the temperature of small particles, typically atoms or ions. Depending on the mechanism used, the temperature achieved can be in the millikelvin, microkelvin, or even nanokelvin regime.

Working:

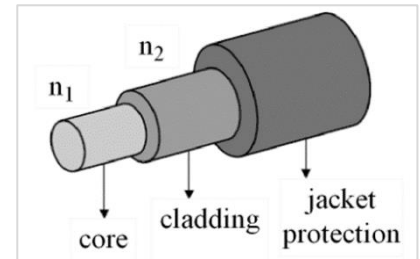
- If an atom is traveling toward a laser beam and absorbs a photon from the laser, it will be slowed by the fact that the photon has momentum; $p = \frac{E}{c} = \frac{h}{\lambda}$.
- It would take a large number of such absorptions to cool the sodium atoms to near 0 K.
- Laser cooling uses a wide range of methods such as (i) Doppler cooling, (ii) Sisyphus cooling, (iii) resolved sideband cooling, (iv) Raman sideband cooling,, etc



Optical Fibers

Optical Fiber is a thin, transparent, cylindrical flexible fiber used for transmitting information from one point to the other. It works on the principle of total internal reflection of light.

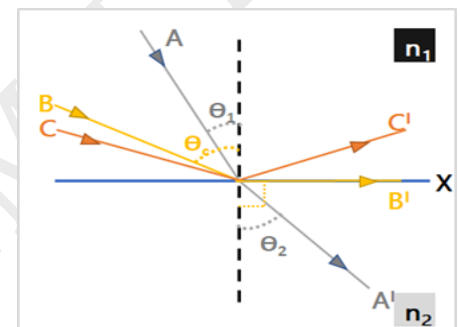
Structure: Optical fiber mainly consists of **core**, **cladding** and **sheath/jacket**. The core is the inner part of fiber made up of a solid dielectric material surrounded by another dielectric material called Cladding. The cladding is enclosed with polyurethane jacket called buffer jacket/sheath, which safeguard the fiber from chemical reaction with surroundings, chemical abrasion, crushing and also gives mechanical strength.



Principle: It works on the principle of total internal reflection.

Total Internal Reflection: is the phenomenon in which the light reflects back into same medium for angles greater than critical angle.

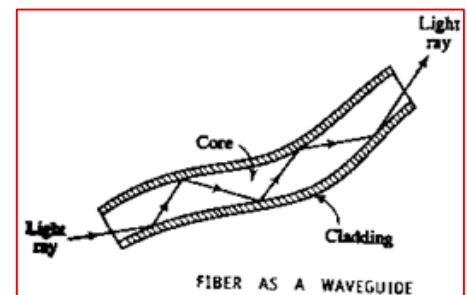
Critical angle: it is the angle of incidence in the medium, for which the angle of refraction is 90. i.e., refracted ray grazes the refracting surface



Propagation of Light through Optical Fiber:

An optical fiber is essentially a **waveguide** for light.

A **waveguide** is a tubular structure through which light could be guided in the form of waves. Since light waves can be guided through a fiber, thus it is called fiber waveguide.



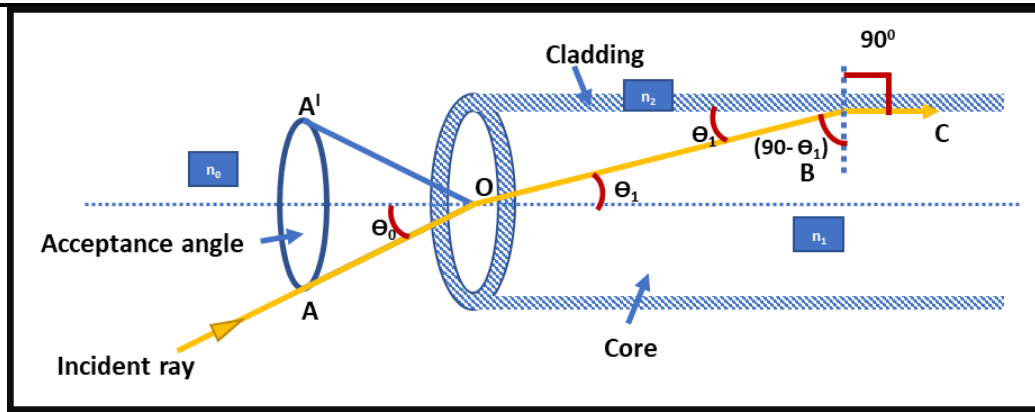
Once light ray enters into core, it propagates by means of multiple total internal reflection at core-cladding interface.

In any optical fiber, the refractive index of cladding is always lesser than that of core. The light signal, which enters into the core, can strike the interface of the core and cladding only at large angles of incidence. The signal undergoes reflection at the points wherever it is incident on the interface. Since each reflection is a total internal reflection, the signal sustains its strength and also confines itself completely within the core during propagation. Thus, the optical fiber functions as a waveguide.

Angle of Acceptance and Numerical Aperture (NA):

Acceptance angle is the semi vertical angle of the imaginary cone within which if the rays enter the fiber, will undergo total internal reflection. Sine of this angle is known as **numerical aperture** of the fiber, which is a measure amount light propagating.

i.e., $NA = \sin \theta_0$ | where θ_0 is the acceptance angle



Let us consider an optical fiber as shown in Fig. Let n_0 , n_1 and n_2 be the refractive indices of surrounding medium, core and cladding of fiber, respectively.

Let a ray AO enters the core at an angle θ_0 and refracted along OB at an angle θ_1 . Further, it proceeds to fall at critical angle of incidence ($\theta_c = 90 - \theta_1$) at B. Because of this critical angle, the ray grazes along BC. It is clear from the Fig that any ray enters with $\theta_0 < \theta_c$ undergoes total internal reflection. Thus, θ_0 is called acceptance angle and $(\sin \theta_0)$ is called numerical aperture, which represents light gathering capability of optical fiber.

Now, by applying the Snell's law at the point of entry 'O',

$$n_0 \sin \theta_0 = n_1 \sin \theta_1 \quad (1)$$

Similarly, by applying the Snell's law at the point B, we have,

$$n_1 \sin(90 - \theta_1) = n_2 \sin 90 \quad \because \angle i = 90 - \theta_1 \text{ and } \angle r = 90$$

$$\Rightarrow n_1 \cos \theta_1 = n_2 \Rightarrow \cos \theta_1 = \frac{n_2}{n_1} \quad (2)$$

$$\text{Rewriting eqn. (1), we have, } \sin \theta_0 = \frac{n_1}{n_0} \sin \theta_1 = \frac{n_1}{n_0} \sqrt{(1 - \cos^2 \theta_1)}$$

$$\text{Substituting for } \cos \theta_1 \text{ from eqn. (2), we have, } \sin \theta_0 = \frac{n_1}{n_0} \sqrt{\left(1 - \frac{n_2^2}{n_1^2}\right)}$$

$$\text{i.e., } NA = \frac{\sqrt{n_1^2 - n_2^2}}{n_0}$$

If the medium surrounding the fiber is air then $n_0 = 1$.

$$\therefore \text{ the above eqn. becomes, } NA = \sqrt{n_1^2 - n_2^2}$$

This is the required expression for numerical aperture (NA).

Modes of propagation:

Defⁿ: The paths along which the light is guided in the fiber, are called modes of propagation.

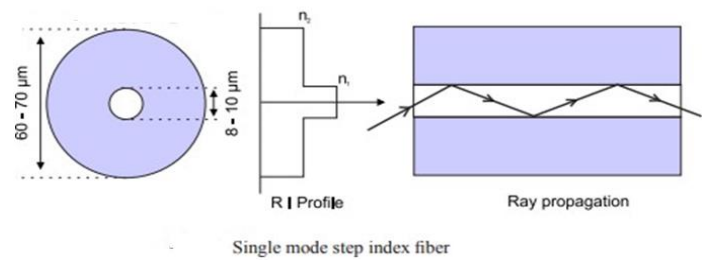
- If Light guided in single path is called single-mode propagation.
- If Light guided in several paths are called multi-mode propagation.

Types of Optical Fibers:

The optical fibers are mainly classified into three categories based on refractive index and number of modes that the fiber can guide.

(1) Step Index Single Mode Fiber:

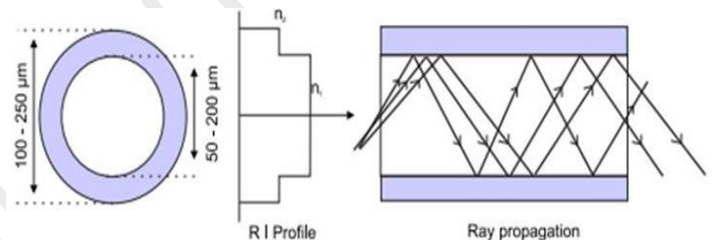
- It has a core and cladding material of uniform refractive index (but $n_{\text{core}} > n_{\text{cladding}}$). This results in a sudden increase in the value of refractive index from cladding to core. Thus, its RI profile takes the shape of a **step**.



- The core diameter value is about **8 to 10 μm** and cladding diameter is 60 to 70 μm .
- Due to its narrow core, it can guide just a single mode. Hence, it is called **single mode fiber**.
- They need lasers as the source of light.
- They find particular application in submarine cable system.

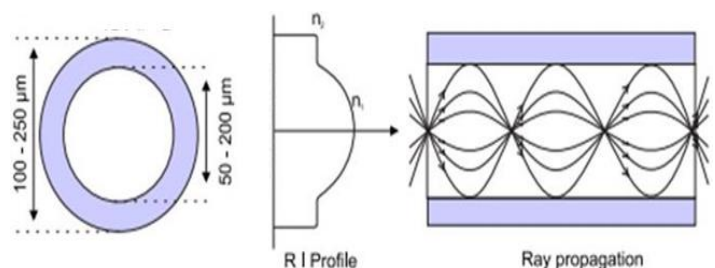
(2) Step Index Multimode Fiber:

- Core of Step index multimode fiber has a much larger diameter by the virtue of which it will support propagation of large number of modes. Hence, it is called **multi-mode fiber**.
- The Core diameter value is **50-200 μm** and cladding diameter is 100-200 μm . Its RI profile is same as that of single mode fiber but larger diameter for core.
- It can accept either a laser or an LED as source of light.
- It's typical application is in data links which has lower bandwidth requirements.



(3) Graded Index Multimode Fiber:

- Here, the core material has a special feature that its refractive index value decreases radially outward direction from the axis and becomes equal to that of cladding at the interface. But the cladding refractive index remains uniform.
- It has core diameter **50-200 μm** of variable refractive index and cladding diameter 100-250 μm .
- Either laser or LED can be the source of light.
- It's typical application is in the telephone trunk between central offices.



Attenuation (Fiber Loss):

Attenuation is the loss of power suffered by the optical signal as it propagates through the fiber. It is also called the fiber loss. It is measured in terms of dB/km.

Attenuation coefficient: When light travels in a material medium, there will always be a loss in its intensity with distance travelled. The loss of intensity is expressed in terms of a quantity called **attenuation coefficient** and denoted as:

$$\alpha = -\frac{10}{L} \log_{10} \left(\frac{P_{out}}{P_{in}} \right) \text{ dB/km}$$

where, L is the length of the fiber (km), P_{in} is the initial intensity with which the light is launched into the fiber (W) and P_{out} is the intensity of the signal received as the output at other end of the fiber (W).

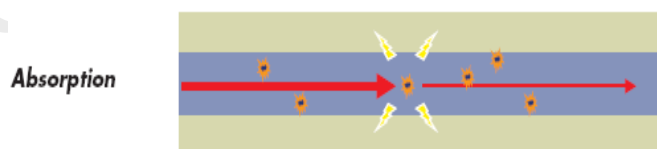
Types of Fiber Losses: The three mechanisms through which attenuation/fiber losses take place, namely ;

- (a) Absorption,
- (b) Scattering and
- (c) Geometrical losses.

(a) Absorption losses: The loss of signal power occurs due to absorption of photons associated with the signal.

Photons are mainly absorbed by, (i) impurities present in the fiber and (ii) intrinsic absorption by the material itself

(i) **Impurities present in the fiber:** During signal propagation, the photons interact with the impurities (such as iron, chromium, cobalt, etc.) present in the fiber, resulting in their absorption and hence a loss. In addition, the other impurity, which would cause significant loss, is the OH ion, which enters into the fiber constitution at the time of fiber fabrication.

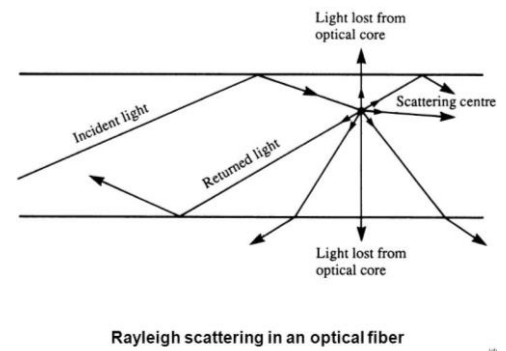


(ii) **Intrinsic absorption:** are the losses due to the material itself, which decreases with the increase of wavelength. The intrinsic losses are insignificant.

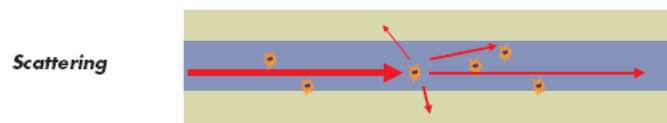
(b) Scattering losses: mainly occurs due to

- (i) Rayleigh scattering and
- (ii) others

(i) Rayleigh scattering: The photons may be scattered because of sharp changes in the refractive index values inside the fiber (see **Fig**). The sharp changes are induced by localized structural inhomogeneity, which is set into fiber during its solidification. This type of scattering is same as the Rayleigh scattering. It occurs whenever photos hits scattering centers whose dimensions are smaller than wavelength. Hence, Rayleigh scattering has a characteristic λ^{-4} dependence, it increases enormously with decreasing wavelength.



(ii) Others: There are defects in the fiber in the form of trapped gas bubbles, un-reacted starting materials and some crystallized region in the glass cause the loss, but this loss can be reduced.



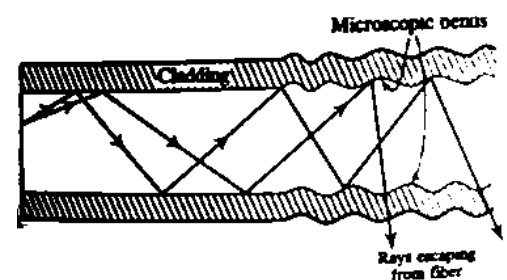
(c) Geometrical loss: occurs due to (i) Macroscopic and
(ii) Microscopic bends

(i) Macroscopic bends: They are the bends with radii much larger compared to fiber diameter. These bends occur while wrapping the fiber on a spool or turning it around a corner. This loss will be negligible for small bends but increases rapidly until the bending reaches a certain critical value.



(ii) Microscopic bends:

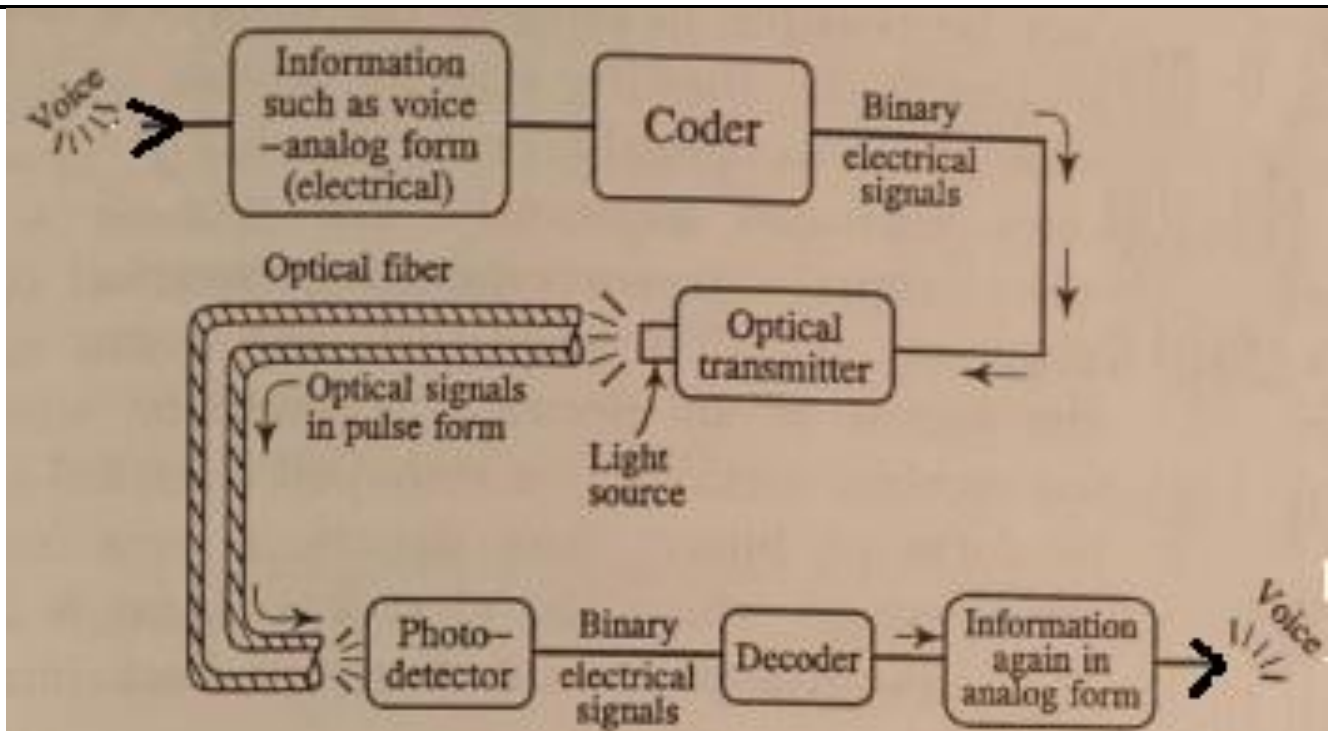
These types of bends are repetitive small-scale fluctuations in the linearity of the fiber axis. They occur due to non-uniformities in the manufacturing of the fiber or by nonuniform lateral pressures created during the cabling of the fiber. This loss could be minimized by using compressible jacket over the fiber.



Application of Optical Fiber:

(a) Communication system (Point-to-point):

Optical fiber communication is the transmission of information using optical signal through optical fiber from one end to other. Thus, it is called as point-to-point communication and its communication mechanism is explained below.



- In point-to-point communication, the information in the form of voice/picture/text, etc. is converted to analog (electrical) signals through the **transducers**.
- The analog signal is converted into binary data (digital form) with the help of **coder**.
- The binary data in the form of electrical pulses are converted into **optical signals** of optical power using **optical transmitter**.
- This **optical signal** is fed to the optical fiber. Only those modes within the angle of acceptance cone will be sustained for propagation.
- At the receiving end of the fiber, the optical signal is fed into a **photo detector**.
- The signal is converted to binary electrical signals by a photo diode.
- **Decoder** converts the sequence of binary electrical signals into an analog signal, which is the same information which was at the transmitting end.

(b) Fiber optic Networking:

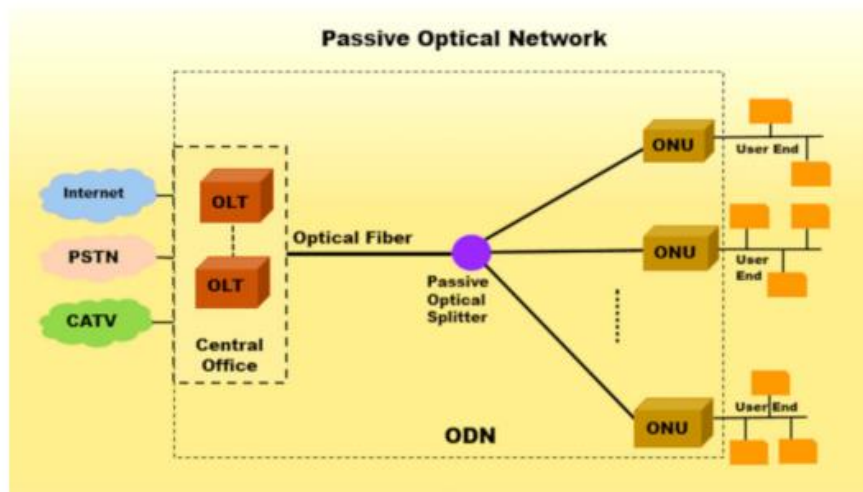
(i) Local Area Network: A Local Area Network (LAN) is a type of computer network that interconnects multiple computers and computer driven devices in a particular physical location. Traditionally copper coaxial cables are used for LAN.

(ii) Passive Optical Network:

Construction and Working:

- A typical Passive Optical Network is comprised of optical line terminations (OLTs), optical splitter, and ONUs (optical network units),

- Generally, an OLT is located at the central office of the server provider, with as many as ONUs situated close to the end users.
- Passive here refers to the unpowered condition of the fiber and splitting/combining components.
- A Passive Optical Network uses optical splitters to separate signals as they progress through the network, sharing strands of fiber optics for different parts of network architecture.
- Data like; Internet, public switched telephone network (PSTN), Cable TV (CATV) etc., can be transmitted from the OLT to ONU using passive optical splitter.



- Advantages:** (i) High speeds and bandwidth.
(ii) Longer distances are possible
(iii) Less chance of errors

Merits of Optical Fibers:

- They can carry large amount of information.
- Materials used for optical fiber are easily available. Hence, *Cost is low*.
- Lightweight and compact in nature, hence they are much *easier to transport*.
- *Not affected by lightning and sparking*
- There is no interference between the communication channels & others. Also, *purity of signals are unaffected*.
- There is no energy radiation from the fiber. Hence, the *loss of information by radiation is ruled out*.
- Since the signals are optical, *no sparks are generated*. ∴ it is protected from corrosive & flammable environments.
- Because of its superior attenuation characteristics, optical fibers support *signal transmission over long distance*.

Demerits of Optical Fibers:

- The optical connectors which are used to connect two fibers, are *highly expensive*.
- If splicing is not done precisely, then the *signal loss will be more*.
- Whenever there is line break, operations required to establish connection are *highly skillful and time consuming*.
- They undergo expansion and contraction with temperature that upset the alignments, which will lead to *line break*.
- *Maintenance costs are high*.

-----@END@-----

Solved Examples (Lasers)

$$\therefore \lambda = 632 \text{ nm}$$

Example 1. A medium in thermal equilibrium at temperature 300 K has two energy levels with a wavelength separation of 1 μm . Find the ratio of population densities of the upper and lower levels.

Data: Temperature, $T = 300 \text{ K}$,

Wavelength separation, $\lambda = 10^{-6} \text{ m}$

To find: Ratio of population densities, $N_2/N_1 = ?$

Solution: We have the Boltzmann factor,

$$\begin{aligned} \frac{N_2}{N_1} &= e^{-\Delta E/KT} = e^{-hc/\lambda KT} = e^{-\left(\frac{hc}{K}\right)\left(\frac{1}{\lambda T}\right)} \\ &= e^{-0.014413\left(\frac{1}{\lambda T}\right)} \\ &= e^{-0.014413/10^{-6} \times 300} \\ &= e^{-48.043} \end{aligned}$$

$$\therefore \frac{N_2}{N_1} = 1.365 \times 10^{-21}$$

Example 2. The ratio of population of two energy levels is 1.059×10^{-30} . Find the wavelength of light emitted by spontaneous emissions at 330 K.

Data: Ratio of population;

$$(N_2/N_1) = 1.059 \times 10^{-30}$$

Temperature, $T = 330 \text{ K}$

To find: The wavelength of light emitted, $\lambda = ?$

Solution: We have the Boltzmann factor,

$$\frac{N_2}{N_1} = e^{-\Delta E/KT}$$

By taking natural log on both sides, we have,

$$\ln\left(\frac{N_2}{N_1}\right) = -\frac{\Delta E}{KT} = -\frac{hc}{\lambda KT}$$

$$\ln(1.059 \times 10^{-30}) = -\frac{6.63 \times 10^{-34} \times 3 \times 10^8}{\lambda \times 1.38 \times 10^{-23} \times 330}$$

$$\therefore \lambda = -\frac{6.63 \times 10^{-34} \times 3 \times 10^8}{\ln(1.059 \times 10^{-30}) \times 1.38 \times 10^{-23} \times 330}$$

Example 3. A laser operating at 632.8 nm emits 3.182×10^{16} photons per second. Calculate the output power of the laser if the input power is 100 watt. Also, find the percentage power converted into coherent light energy.

Data:

Wavelength of the laser light, $\lambda = 632.8 \times 10^{-9} \text{ m}$

No. of photons emitted per second, $n = 3.182 \times 10^{16}$

Input power, $P_{in} = 100 \text{ W}$

To find: Percentage of power converted into laser light energy, (i.e., percentage power efficiency), $P_{out} = ?$

Solution: We have, energy of each photon,

$$E = \frac{hc}{\lambda} = \frac{6.623 \times 10^{-34} \times 3 \times 10^8}{632.8 \times 10^{-9}}$$

$$E = 3.143 \times 10^{-19} \text{ J}$$

But, $P_{out} = \text{Total Energy emitted/second}$

i.e., $P_{out} = [\text{No. of photons emitted/second}] \times E$

$$P_{out} = n \times E = 3.182 \times 10^{16} \times 3.143 \times 10^{-19}$$

$$\therefore P_{out} = 0.01 \text{ W.}$$

$$\% \text{ Power efficiency} = \frac{P_{out}}{P_{in}} \times 100$$

$$\% \text{ Power efficiency} = \frac{0.01}{100} \times 100 = 0.01$$

Example 4. The average output power of laser source emitting a laser beam of wavelength 632.8 nm is 5 mW. Find the number of photons emitted per second by the laser source.

Data: Wavelength of the emitted light,

$$\lambda = 632.8 \text{ nm} = 632.8 \times 10^{-9} \text{ m}$$

$$P_{out} = 5 \text{ mW} = 5 \times 10^{-3} \text{ W}$$

To find: No. of photons emitted/second, $n = ?$

Solution: We know that the energy difference,

$$\Delta E = h\nu = \frac{hc}{\lambda} = \frac{6.623 \times 10^{-34} \times 3 \times 10^8}{632.8 \times 10^{-9}}$$

$$\Delta E = 3.143 \times 10^{-19} \text{ J}$$

This energy difference becomes the energy of each of the emitted photon. If n is the number of photons emitted per second to give a power output of 5 mW, then,

$$n \times \Delta E = 5 \text{ mW} = 5 \times 10^{-3} \text{ J/s}$$

$$\text{i.e., } n = \frac{5 \times 10^{-3}}{3.143 \times 10^{-19}}$$

$$\therefore n = 1.59 \times 10^{16}$$

Example 5. Calculate the number of photons emitted per pulse of duration 1 ms given the power output of LASER 3 mW and the wavelength of laser 632.8 nm.

Data: Wavelength of the emitted light,
 $\lambda = 632.8 \text{ nm} = 632.8 \times 10^{-9} \text{ m}$

$$P_{\text{out}} = 3 \text{ mW} = 3 \times 10^{-3} \text{ W}$$

To find: No. of photons emitted/second, $n = ?$

Solution: We know that the energy difference,

$$\Delta E = h\nu = \frac{hc}{\lambda} = \frac{6.623 \times 10^{-34} \times 3 \times 10^8}{632.8 \times 10^{-9}}$$

$$\Delta E = 3.143 \times 10^{-19} \text{ J}$$

$$\text{Given, the energy emitted per ms by laser} = \frac{3 \text{ mW}}{1 \times 10^{-3}} = 3 \text{ J}$$

Thus the number of photons emitted per second

$$= \frac{3}{3.14 \times 10^{-19}} = 0.96 \times 10^{19}$$

$$\text{i.e., } n = 0.96 \times 10^{19}$$

Solved Examples (Optical Fibers)

Example 1. The refractive indices of core and cladding are 1.50 and 1.48, respectively, in an optical fiber. Find the numerical aperture and angle acceptance.

Data: Refractive index of the core, $n_1 = 1.50$

Refractive index of the cladding, $n_2 = 1.48$

To find: (a) Numerical Aperture, $NA = ?$

(b) Angle of acceptance $\theta_0 = ?$

Solution: We have the relation for NA as

$$NA = \sqrt{(n_1^2 - n_2^2)} = \sqrt{(1.50^2 - 1.48^2)}$$

$$NA = 0.2440$$

The angle of acceptance θ_0 is related to NA as,

$$\theta_0 = \sin^{-1}(NA) = \sin^{-1}(0.2440)$$

$$\theta_0 = 14.1^\circ$$

Example 2. An optical fiber has core material with refractive index 1.55, and its cladding material has refractive index of 1.50. The light is launched into it in air. Calculate the numerical aperture and the acceptance angle.

Data: Refractive index of the core, $n_1 = 1.55$

Refractive index of the cladding, $n_2 = 1.50$

Surrounding medium is air i.e., $n_0 = 1$

To find: (a) Numerical Aperture, $NA = ?$

(b) Angle of acceptance $\theta_0 = ?$

Solution: (a) For the case of surrounding medium as air ($n_0 = 1$), the NA is given by;

$$NA = \sqrt{(n_1^2 - n_2^2)} = \sqrt{(1.55^2 - 1.50^2)}$$

$$NA = 0.3930$$

(b) The angle of acceptance is related to NA as,

$$\theta_0 = \sin^{-1}(N.A.) = \sin^{-1}(0.3930)$$

$$\theta_0 = 23^\circ$$

Example 3. The N.A. of an optical fiber is 0.2 when surrounded by air. Determine the RI of its core given the RI of cladding is 1.59. Also find the acceptance angle when the fiber is in water. Assume RI of water is 1.33.

Data: N.A. while fiber is in air $(N.A.)_{air} = 0.2$,

Refractive index of the water, $n_0 = 1.33$,

Refractive index of the cladding, $n_2 = 1.59$

To find: (a) Refractive index of the core, $n_1 = ?$

(b) Acceptance angle while the fiber is in water $\theta_0 = ?$

Solution: (a) While in air N.A. is given as;

$$N.A. = \sqrt{(n_1^2 - n_2^2)}$$

$$\text{i.e., } 0.2 = \sqrt{(n_1^2 - 1.59^2)}$$

Squaring on both side

$$0.04 = n_1^2 - 2.581$$

$$\rightarrow n_1 = 1.6$$

(b) While in water N.A. is given as;

$$N.A. = \frac{\sqrt{(n_1^2 - n_2^2)}}{n_0} = \frac{\sqrt{(1.6^2 - 1.59^2)}}{1.33}$$

$$N.A. = 0.1504$$

But, the angle of acceptance is related to NA as,

$$\theta_0 = \sin^{-1}(N.A.) = \sin^{-1}(0.1504)$$

$$\theta_0 = 8.65^\circ$$

Example 4. The angle of acceptance of an optical fiber is 30° when kept in air. Find the angle of acceptance when it is in the medium of refractive index 1.33.

Data: Angle of acceptance in air, $\theta_0 = 30^\circ$,

RI of refracting medium, $n_0 = 1.33$,

To find: Angle of acceptance when the fiber is in the refracting medium, $\theta_0 = ?$

Solution: We have the equation

$$\sin \theta_0 = \frac{\sqrt{(n_1^2 - n_2^2)}}{n_0}$$

When the surrounding medium is air, then

$$n_0 = 1, \text{ and } \theta_0 = 30^\circ$$

$$\therefore \sin 30 = \frac{\sqrt{(n_1^2 - n_2^2)}}{1}$$

$$\rightarrow 0.5 = \sqrt{(n_1^2 - n_2^2)} \quad \dots\dots (1)$$

When the surrounding medium is of refractive index 1.33 let θ_0 be the acceptance angle

$$\sin \theta_0 = \frac{\sqrt{(n_1^2 - n_2^2)}}{n_0} \rightarrow \sin \theta_0 = \frac{\sqrt{(n_1^2 - n_2^2)}}{1.33}$$

$$\text{Or, } 1.33 \sin \theta_0 = \sqrt{(n_1^2 - n_2^2)} \quad \dots\dots (2)$$

from Eqs (1) and (2), we get

$$1.33 \sin \theta_0 = 0.5 \rightarrow \theta_0 = \sin^{-1}\left(\frac{0.5}{1.33}\right)$$

$$\theta_0 = 22^\circ$$

The acceptance angle when the fiber is in the medium of refractive index 1.33 is 22° .

Example 5. The attenuation in an optical fiber is 3.6 dB/km. what fraction of its initial intensity remains after i) 1 km, ii) after 3km?

Data: Attenuation in the fiber, $\alpha = 3.6$ dB/km.

To find:

$$\text{i) After 1 km distance, } \left(\frac{P_{in}}{P_{out}}\right) = ?$$

$$\text{ii) After 3 km distance, } \left(\frac{P_{in}}{P_{out}}\right) = ?$$

Solution: We have,

$$\alpha = -\frac{10}{L} \log_{10} \left(\frac{P_{out}}{P_{in}} \right)$$

Where L is length of the fiber is expressed in kms.

$$\log_{10} \left(\frac{P_{out}}{P_{in}} \right) = -\frac{\alpha L}{10}$$

$$\left(\frac{P_{out}}{P_{in}} \right) = 10^{-\left(\frac{\alpha L}{10}\right)}$$

∴ We have, at the end of 1 km distance, $L=1$.

$$\therefore \left(\frac{P_{out}}{P_{in}} \right) = 10^{-\left(\frac{3.6}{10}\right)} = \mathbf{0.436}.$$

At the end of 3 kms, $L=3$.

$$\therefore \left(\frac{P_{out}}{P_{in}} \right) = 10^{-\left(\frac{3 \times 3.6}{10}\right)} = \mathbf{0.0832}.$$

Therefore, at the end of 1 km and 3 kms, the fraction of intensity of light that is available will be 0.436 (i.e. 43.6%) and 0.0832 (i.e. 8.3%) times that of the intensity of light that was launched into the fiber intensity.

Example 7. Estimate the attenuation in an optical fiber of length 500 m, when a light signal of power 100 mW emerges out of the fiber with a power 90 mW.

Data: Length of the optical fiber, $L = 500 \text{ m} = 0.5 \text{ km}$

Input power of signal, $P_{in} = 100 \text{ mW} = 100 \times 10^{-3} \text{ W}$

Output power of signal, $P_{out} = 90 \text{ mW} = 90 \times 10^{-3} \text{ W}$

To find: Fiber attenuation, $\alpha = ?$

Solution: The fiber attenuation α is given by,

$$\alpha = -\frac{10}{L} \log_{10} \left(\frac{P_{out}}{P_{in}} \right) \text{ dB/km},$$

$$\alpha = -\frac{10}{L} \log_{10} \left(\frac{90}{100} \right)$$

$$\alpha = \mathbf{0.915 \text{ dB/km}}$$

Example 8. Compare the acceptance angle of an optical fiber placed in air and water given the RI of water 1.33 and RI of core and clad 1.5 and 1.45 respectively.

Data:

RI of Core $n_1 = 1.5$; RI of Clad $n_2 = 1.45$; RI of water = 1.33

Solution: W.K.T,

For optical fiber placed in air;

$$\theta_0 = \sin^{-1}(NA) = \sin^{-1}(\sqrt{n_1^2 - n_2^2})$$

$$= \sin^{-1}(\sqrt{1.5^2 - 1.45^2})$$

$$\therefore (\theta_0)_{air} = \mathbf{22.58^\circ}$$

For optical fiber placed in water medium;

$$\theta_0 = \sin^{-1}(NA) = \sin^{-1} \left(\frac{\sqrt{n_1^2 - n_2^2}}{n_0} \right) = \sin^{-1} \left(\frac{\sqrt{1.5^2 - 1.45^2}}{1.33} \right)$$

$$\therefore (\theta_0)_{water} = \mathbf{16.78^\circ}$$

The acceptance angle of optical fiber placed in air (22.58°) is greater than the one which is placed in water medium (16.78°).

